CATEGORY-SELECTIVE BRAIN NETWORKS WITH GRAPH NEURAL NETWORKS

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ABSTRACT

We study how distributed functional coupling encodes visual categories in the human cortex. Using high-resolution 7T fMRI from the Natural Scenes Dataset (NSD), we construct parcel-level connectivity graphs and train a *Signed* Graph Neural Network (GNN) that models both positive and negative interactions. Interpretability is embedded through a shared, sparse edge mask and complemented by class-specific Gradient×Input saliency. The model achieves accurate multi-class decoding (*sports*, *food*, *vehicle*) while revealing reproducible, biologically meaningful subnetworks within ventral and dorsal pathways. This framework bridges machine learning and neuroscience by extending voxel-level findings on food selectivity to a connectivity-selective representation of visual processing.

Index Terms— fMRI, Natural Scenes Dataset, Graph Neural Networks, Explainability, Functional Connectivity

1. INTRODUCTION

Functional Magnetic Resonance Imaging (fMRI) provides a non-invasive means to study human brain function by measuring blood-oxygen-level-dependent (BOLD) signals across distributed cortical regions. Modeling these signals as networks enables characterization of how distinct brain areas interact during perception. Recent advances in deep learning, particularly Graph Neural Networks (GNNs), offer powerful tools for analyzing such brain connectivity patterns, as they naturally capture the graph-structured organization of neural interactions.

The application of GNNs to fMRI has shown promise in decoding cognitive states, disease phenotypes, and functional organization by directly leveraging the topology of brain graphs [1]. However, most existing works focus on resting-state connectivity or small-scale datasets, and relatively few address the decoding of *stimulus-driven* activity during naturalistic perception. Moreover, interpretability remains a central challenge: while GNNs can achieve high decoding accuracy, the underlying decision mechanisms often remain fuzzy. To bridge this gap, recent efforts have introduced explainable-AI (XAI) strategies—such as edge masking, contrastive learning, and saliency-based attribution—to identify which connections contribute most to classification outcomes, though these approaches have rarely been applied to large-scale, high-resolution datasets like NSD.

Here, we develop an interpretable *Signed Graph Neural Network (Signed GNN)* to decode functional connectivity patterns evoked by naturalistic visual stimuli. Using the Natural Scenes Dataset (NSD) [2]—a 7 Tesla fMRI resource comprising thousands of COCO images—we classify stimulus-evoked responses into three semantic categories: *sports*, *food*, and *vehicle*. Each image's category is derived through a multimodal fusion of COCO 2017 instance masks and textual captions, combining object-level and linguistic evidence into unified "super-category" representations.

Our model jointly learns from positive and negative functional connections, representing cooperative and competitive neural interactions, respectively. Intrinsic interpretability is embedded through a sparsity-regularized shared edge mask that identifies globally relevant connections, while Gradient×Input saliency captures class-specific patterns. This combination provides both global and category-specific insights into how connectivity supports perception.

From a neuroscientific perspective, prior studies have identified selective responses to food in the ventral visual pathway [3, 4, 5], emphasizing fusiform and parahippocampal regions. However, these findings primarily describe local voxel activations. The present approach extends this understanding to the *connectivity domain*, revealing how distributed networks—spanning ventral, dorsal, and limbic systems—cooperate during category-specific processing.

This work advances the use of explainable graph learning for stimulus-driven fMRI decoding by integrating interpretable GNN mechanisms with large-scale, naturalistic data. The resulting framework provides biologically meaningful insights into the functional architecture of category-selective networks in the human brain.

2. METHODS

2.1. Dataset

We used the Natural Scenes Dataset (NSD) [2], a large-scale 7T fMRI resource that captures brain responses to thousands of naturalistic images. Eight participants viewed up to 10,000 color images from the COCO (*Common Objects in Context*) dataset, which contains everyday scenes with multiple objects and complex backgrounds. Each image was presented for 4 s while participants performed a continuous rating task, in which they reported whether the current image had been presented at any previous point in the experiment. This behavioral engagement ensured reliable attention and reduced habituation across the long experimental sessions. The NSD data are acquired at submillimeter resolution and mapped onto individual cortical surfaces, allowing high-fidelity analyses of fine-scale functional patterns.

2.2. Preprocessing and Parcellation

We used right-hemisphere fMRI responses that were preprocessed following the NSD pipeline, including motion correction, trend removal, and cortical surface projection. Each cortical surface was parcellated into 180 regions of interest (ROIs) using the multimodal HCP-MMP1 atlas [6], which integrates structural, functional, and connectivity information to define consistent cortical boundaries. For each subject, we extracted parcel-averaged time series across all runs, producing standardized, noise-reduced signals representative of regional neural activity.

Some author footnote.

Fig. 1: Example of a cropped NSD stimulus with COCO instance masks (cars, tennis, people) and corresponding captions, for example: "a man who is playing tennis on a tennis court", Both mask- and text-based features contribute to category fusion for *sports*.

Each NSD image was associated with its COCO 2017 instance annotations and textual captions to obtain a semantic description of the visual content. To study category-selective responses, we defined three super-categories—*sports*, *food*, and *vehicle*—and computed their relative evidence per image from two complementary modalities: (i) the fraction of each object mask covering the cropped NSD stimulus, and (ii) the normalized frequency of category-specific keywords appearing in the captions. These two evidence sources were fused using a weighted rule,

$$\text{score}_c = \alpha_c \, \text{text}_c + (1 - \alpha_c) \, \text{mask}_c,$$

where the weight α_c was category-dependent (e.g., *sports* relied more on text, *vehicle* more on mask area). An example of this multimodal fusion is shown in Fig. 1. Images were assigned to the super-category base on their score, as each 10% in the score representing a repetition of an image in the particular category, and a global train/validation/test split was created before any fMRI processing to avoid data leakage. A uniform 10% rounding rule was applied to maintain balanced sample counts across categories.

2.3. Connectivity Construction

For each subject, fMRI time series were arranged as tensors of shape (N, P, T), where N is the number of stimulus trials, P=180 the number of parcels, and T the number of timepoints per trial. Trials were grouped into shuffled blocks of K=60 and concatenated along the temporal dimension. Parcel signals were z-scored within each block, and Pearson correlations were computed between all parcel pairs, yielding symmetric connectivity matrices of size $(P \times P)$. Diagonal elements were fixed to 1, and correlation values were Fisher-clipped to the range $[-1, 1]$. These matrices served as the input graphs for GNN training, separately constructed and balanced across the three super-categories for subsequent classification and explainability analyses.

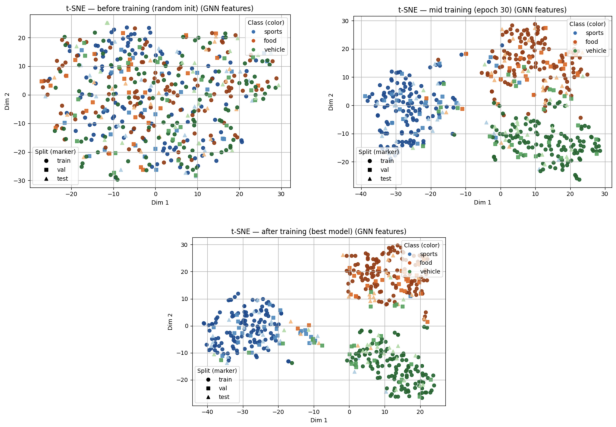

Fig. 2: t-SNE progression of learned graph embeddings across training stages: (a) random initialization, (b) mid-training (epoch 30), and (c) after convergence. Each cluster represents the learned embedding space for the three visual categories (sports, food, and vehicle).

2.4. Signed GNN Architecture and Training

Each sample is represented by a two-channel adjacency tensor (A^+, A^-) capturing positive and negative correlations. The Signed-GNN comprises two graph-convolutional layers (128 and 256 units), global mean pooling, and a two-layer classifier. A shared, symmetric edge mask $M \in [0, 1]^{P \times P}$ multiplicatively gates both A^+ and A^- and is regularized to encourage sparsity and near-binary values:

$$\mathcal{L}_{\text{mask}} = \lambda_1 \, \overline{|M|} + \lambda_e \, \overline{M(1 - M)}, \; \lambda_1 = \lambda_e = 3{\times}10^{-3}.$$

Optimization used class-weighted cross-entropy with label smoothing (0.1), the AdamW optimizer, and early stopping based on validation loss.

2.5. Explainability

Two complementary interpretability strategies were applied to understand the learned representations.

(i) Edge mask. The learnable mask M provides an internal measure of edge importance shared across all samples. High-weight entries indicate globally influential connections that consistently contribute to classification. Visualizing M reveals class-agnostic connectivity backbones and enables direct inspection of which cortical links the network relies on.

(ii) Class-specific saliency. To capture category-dependent connectivity, we computed per-class Gradient×Input saliency maps. The resulting $S_c \in \mathbb{R}^{P \times P}$ quantifies each connection's contribution to class c. After symmetrization and removal of the diagonal, we extracted the top-weighted edges and computed node-level relevance by summing incident edge weights. These parcel-wise relevance scores formed interpretable cortical maps for the *sports*, *food*, and *vehicle* categories.

Table 1: Per-subject test accuracy and per-class Average Precision (AP). Macro-AP is the mean of class APs per subject. The last row reports across-subject means.

Subject	Accuracy	Sports	Food	Vehicle	Macro-AP
1	0.790	0.916	0.939	0.829	0.895
2	0.769	0.860	0.930	0.837	0.876
3	0.640	0.739	0.843	0.643	0.742
4	0.870	0.923	0.942	0.880	0.915
5	0.810	0.929	0.949	0.918	0.932
6	0.818	0.950	0.959	0.910	0.940
7	0.790	0.854	0.961	0.804	0.873
8	0.746	0.870	0.859	0.858	0.863
Mean	**0.78**	**0.88**	**0.92**	**0.84**	**0.88**

3. RESULTS AND DISCUSSION

3.1. t-SNE Visualization of Graph Embeddings

The t-SNE embeddings (Fig. 2) exhibited a consistent trajectory across all eight participants: (i) strong category overlap before training, (ii) emerging separation by epoch 30, and (iii) clearly segregated clusters at convergence. This reproducible evolution suggests that the Signed GNN progressively organizes the latent graph manifold into category-selective subspaces. Notably, intermediate stages revealed partial overlap between *sports* and *vehicle* embeddings, which later disentangled, reflecting hierarchical refinement of learned connectivity features. The similarity of trajectories across subjects indicates that training converges toward shared representational geometry rather than subject-specific idiosyncrasies.

3.2. Quantitative Evaluation

Training converged within 50–60 epochs with stable validation losses and minimal overfitting. Across subjects, the Signed GNN achieved an average test accuracy of **0.78** and macro-average precision (AP) of **0.88** (Table 1). Performance variability correlated moderately with head-motion indices: participants S03 and S08, who showed greater motion, had slightly reduced accuracy. Category-wise, *food* exhibited the highest separability (AP = 0.92), followed by *sports* (AP = 0.88) and *vehicle* (AP = 0.84). The narrow inter-subject standard deviation (±0.06) demonstrates strong reproducibility of classification performance despite individual differences in signal quality.

3.3. Explainability: Category-Specific Connectivity Patterns

Model explanations derived from (i) the intrinsic *edge mask* and (ii) Gradient×Input saliency revealed coherent, biologically interpretable subnetworks (Fig. 3). These networks correspond to canonical functional systems spanning ventral, dorsal, and fronto-limbic territories.

For **food**, high-weight nodes clustered within ventral occipitotemporal cortex (*LO1–2*, *VMV1–2*, *PIT*, *PH*, *FFC*) and extended into limbic–orbitofrontal regions (*FOP4*, *PHA2*). This constellation delineates a ventral–limbic circuit integrating visual, gustatory, and reward-related processes—consistent with reports of food selectivity in ventral visual cortex and orbitofrontal cortex.

For **sports**, salient hubs emerged across the dorsal visuomotor stream—*MT/MST*, *V7*, *VIP/LIP*, and *7Am*—with additional fronto-parietal contributions (*IFJp*, *FEF*, *55b*). This pattern reflects dynamic coupling between occipito-parietal and premotor areas supporting motion perception, attention, and action planning.

For **vehicles**, top-weighted parcels localized to ventral occipitotemporal cortex (*LO1–3*, *VMV1–3*) and posterior parietal regions (*PGs*, *V3B/V3CD*, *POS2*), indicating integrated processing of object form and spatial layout within the ventral–occipital pathway.

Connectivity-Level Selectivity. Across participants, the Signed GNN revealed stable inter-regional motifs linking sensory, associative, and limbic nodes. These patterns provide a network-level view of how functional coupling encodes category information beyond voxel- or ROI-based representations. Together with the high cross-subject correlations (Fig. 4), the results demonstrate that the learned edge-weight structures reflect reproducible, biologically meaningful organization rather than subject-specific noise.

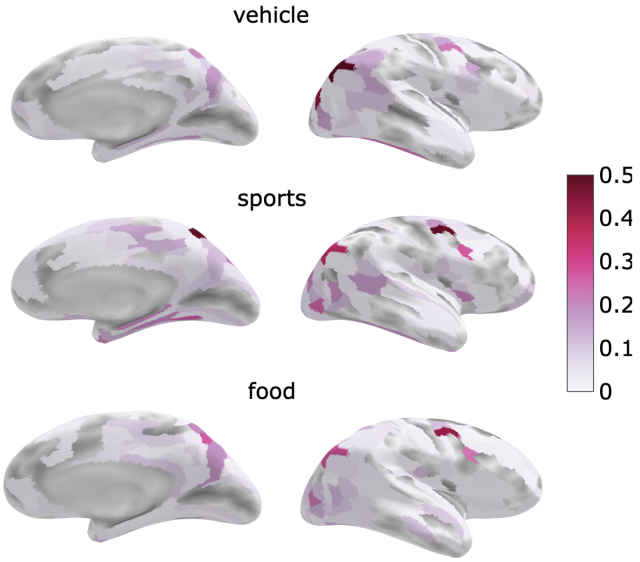

Fig. 3: Top-weighted parcels averaged across eight subjects, showing distinct subnetworks for *sports*, *food*, and *vehicle*.

Table 2: Top recurring HCP-MMP1 parcels across subjects per category (frequency out of 8).

Parcel / Region	Food	Sports	Vehicle
FOP4 (Frontal Operculum)	7	0	0
PHA2 (Parahippocampal)	6	3	5
FFC (Fusiform Complex)	5	2	4
PGs (Parietal Gyrus)	3	6	8
MT/MST (Motion Complex)	2	8	3
V7 (Dorsal Visual)	3	7	7
VIP/LIP (Visuomotor)	2	7	2
7Am (Superior Parietal)	3	7	3
LO1–2 (Lateral Occipital)	4	4	7
VMV1–2 (Ventral Visual)	5	3	7
Overlap (% subjects)	**75.0**	**100.0**	**87.5**

Each category engages a distinct but partially overlapping subnetwork spanning ventral, dorsal, and limbic areas.

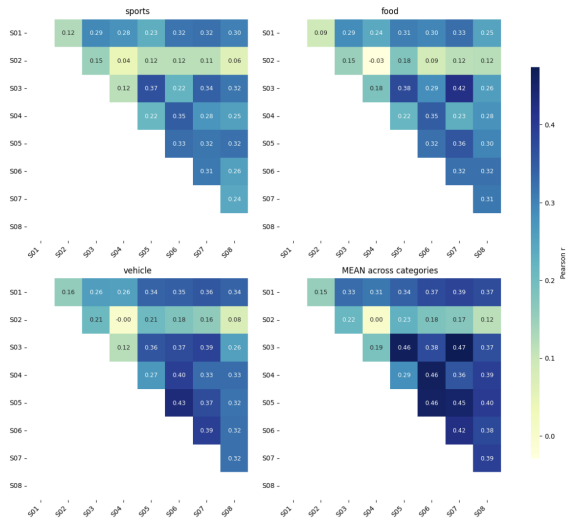

Fig. 4: correlation between the different subjects, for each category and the mean of all

3.4. Cross-Subject and Cross-Study Consistency

Despite strong group convergence, inter-individual variability persisted: participants S03 and S08—characterised by higher motion indices—exhibited reduced ventral–limbic coupling while preserving dorsal activations. This suggests that higher-order associative couplings may be more susceptible to motion/fixation noise than early sensory links.

The subnetworks identified here align closely with prior findings on visual category selectivity. For **food**, ventral occipitotemporal and orbitofrontal nodes overlapped with the food-selective fusiform regions reported by Jain et al. [7] and ventral-temporal activations described by Chen et al. [8] and Lindquist et al. [9]. Our results extend these voxel-level observations by revealing functional coupling between fusiform/VMV parcels and frontal–insular regions (*FOP4*, *PHA2*), supporting multisensory integration of visual and gustatory information.

For **sports** and **vehicles**, the dorsal and ventral subnetworks mirror canonical visual-stream organization. Action-related dorsal activations (*MT/MST*, *V7*, *VIP*) align with visuomotor findings from Ayzenberg et al. [10], whereas ventral-occipital nodes (*LO/VMV*) parallel object-centered representations reported by Andresen et al. [11]. Together, these patterns delineate three reproducible systems: (i) a ventral–limbic network supporting food and reward processing; (ii) a dorsal visuomotor network mediating motion and action perception; and (iii) a ventral–occipital network specialized for object and scene analysis. This tri-axial organization mirrors canonical visual hierarchies and underscores the biological interpretability of the Signed GNN.

3.5. Methodological and Conceptual Implications

From a methodological standpoint, these findings extend interpretable GNN frameworks for brain connectivity [1, 12]. By integrating sparsity-regularized masks with saliency analysis, the model quantifies both global and class-specific edge relevance. This dual interpretability enables comparison across individuals and tasks, revealing consistent cognitive subnetworks in naturalistic perception—an advance over prior applications limited to resting-state or clinical data.

Limitations. First, COCO images often depict multiple objects, introducing label ambiguity despite fusion thresholds. Second, analyses were restricted to the right hemisphere; whole-cortex integration will further clarify inter-hemispheric coordination. Finally, we used static Pearson correlations; dynamic connectivity could refine temporal specificity of category representations.

Overall, the Signed GNN achieved accurate decoding while exposing interpretable, population-consistent connectivity patterns that align with established ventral and dorsal visual hierarchies, bridging machine learning and human neuroscience.

4. CONCLUSION

This study demonstrates that Signed Graph Neural Networks can decode and interpret large-scale fMRI connectivity during natural image viewing. By combining intrinsic edge masks with gradient-based saliency, the model revealed reproducible, category-specific subnetworks for *food*, *sports*, and *vehicle* stimuli that align with known ventral and dorsal visual hierarchies. The approach bridges machine learning and neuroscience by uncovering population-consistent, biologically meaningful patterns of brain connectivity.

Future work will extend this framework to examine individual differences—such as sex-related variations and personal preferences—and to explore whether network-level decoding can illuminate cognitive or learning differences. More broadly, interpretable graph models hold promise for mapping how dynamic brain networks encode semantic and affective meaning in natural perception.

5. REFERENCES

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